

## SANLIURFA GAP, Turkey

WMO: 172710

Lat: 37.445N

Lon: 38.904E

Elev: 2697

StdP: 13.32

Time Zone: 3.00 (E03)

Period: 09-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 25.0       | 28.4 | 9.0                           | 9.6 | 35.0 | 13.4 | 12.0 | 38.4 | 29.1                  | 43.9 | 26.0 | 43.2 | 10.5                  | 20   | 0.517 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 23.7                   | 102.1           | 64.3 | 100.2 | 63.9 | 98.3 | 63.5 | 69.0                | 88.6 | 67.6 | 87.6 | 66.3 | 88.8 | 10.6                 | 280  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 62.7                            | 94.2 | 79.8 | 60.6 | 87.4 | 78.2 | 58.7 | 81.5 | 76.5 | 34.9          | 87.8 | 33.6 | 87.8 | 32.5 | 88.2 | 73.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |       |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|-------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|
| 1%                |      | 2.5% | 5%                         | Mean |       | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |
| Min               |      | Max  |                            | Min  | Max   | Min                | Max                                                | Min       | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  |                            | (e)  | (f)   | (g)                | (h)                                                | (i)       | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 22.6              | 19.6 | 17.0 | DB                         | 20.9 | 106.0 | 4.2                | 2.7                                                | 17.9      | 107.9 | 15.4       | 109.5 | 13.0       | 111.0 | 9.9        | 112.9 |
|                   |      |      | WB                         | 18.7 | 71.2  | 4.2                | 2.1                                                | 15.7      | 72.7  | 13.3       | 74.0  | 10.9       | 75.2  | 7.8        | 76.7  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 62.6  | 39.7  | 43.4  | 49.7  | 58.6  | 67.9  | 78.9  | 85.9  | 86.0  | 77.8  | 66.1  | 53.0  | 43.5 | (6)  |
| (7)  |                                                                                  | DBStd       | 17.15 | 5.43  | 6.00  | 6.21  | 6.09  | 6.04  | 5.20  | 4.00  | 3.59  | 5.87  | 6.19  | 6.03  | 5.46 | (7)  |
| (8)  |                                                                                  | HDD50       | 853   | 320   | 195   | 83    | 7     | 0     | 0     | 0     | 0     | 0     | 0     | 40    | 208  | (8)  |
| (9)  |                                                                                  | HDD65       | 3198  | 785   | 604   | 476   | 203   | 39    | 0     | 0     | 0     | 3     | 63    | 359   | 666  | (9)  |
| (10) |                                                                                  | CDD50       | 5470  | 0     | 11    | 72    | 266   | 555   | 866   | 1112  | 1115  | 834   | 500   | 131   | 8    | (10) |
| (11) |                                                                                  | CDD65       | 2339  | 0     | 0     | 0     | 12    | 129   | 416   | 647   | 650   | 387   | 98    | 0     | 0    | (11) |
| (12) |                                                                                  | CDH74       | 30601 | 0     | 0     | 0     | 103   | 1281  | 5151  | 9564  | 9471  | 4347  | 683   | 1     | 0    | (12) |
| (13) |                                                                                  | CDH80       | 16814 | 0     | 0     | 0     | 7     | 404   | 2662  | 5825  | 5689  | 2082  | 145   | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 8.7   | 9.2   | 8.4   | 9.0   | 8.6   | 8.8   | 10.0  | 9.6   | 8.6   | 8.1   | 7.8   | 7.3   | 8.8  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 17.3  | 3.3   | 2.8   | 2.2   | 1.7   | 1.1   | 0.2   | 0.0   | 0.1   | 0.2   | 1.0   | 2.0   | 2.8  | (15) |
| (16) |                                                                                  | PrecMax     | 32.4  | 7.3   | 6.5   | 10.7  | 4.8   | 4.7   | 1.1   | 0.3   | 0.7   | 2.7   | 2.8   | 6.9   | 7.4  | (16) |
| (17) |                                                                                  | PrecMin     | 11.2  | 0.7   | 0.6   | 0.1   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2  | (17) |
| (18) |                                                                                  | PrecStd     | 4.7   | 1.7   | 1.4   | 2.0   | 0.9   | 1.0   | 0.3   | 0.1   | 0.2   | 0.5   | 0.9   | 1.4   | 1.8  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 55.3  | 64.8  | 69.5  | 80.7  | 91.5  | 100.7 | 105.7 | 104.3 | 98.8  | 87.4  | 73.2  | 65.8 | (19) |
| (20) |                                                                                  | MCWB        | 44.9  | 49.5  | 51.9  | 56.8  | 60.2  | 64.3  | 64.9  | 64.9  | 62.3  | 58.7  | 50.3  | 48.7  | (20) |      |
| (21) |                                                                                  | 2%          | DB    | 51.9  | 59.4  | 66.3  | 76.9  | 87.7  | 98.3  | 102.5 | 102.1 | 95.9  | 83.9  | 69.9  | 57.3 | (21) |
| (22) |                                                                                  | MCWB        | 43.3  | 47.8  | 51.0  | 54.8  | 59.9  | 63.1  | 64.1  | 64.4  | 61.7  | 58.1  | 50.7  | 46.7  | (22) |      |
| (23) |                                                                                  | 5%          | DB    | 49.8  | 55.6  | 64.0  | 74.0  | 84.4  | 95.0  | 100.4 | 100.1 | 93.3  | 80.8  | 66.5  | 53.9 | (23) |
| (24) |                                                                                  | MCWB        | 42.9  | 46.4  | 50.6  | 54.5  | 59.4  | 61.9  | 63.9  | 64.2  | 61.1  | 57.3  | 50.2  | 45.3  | (24) |      |
| (25) |                                                                                  | 10%         | DB    | 47.7  | 53.1  | 60.8  | 71.2  | 81.0  | 91.8  | 98.5  | 98.2  | 90.6  | 77.4  | 64.0  | 51.5 | (25) |
| (26) |                                                                                  | MCWB        | 42.2  | 45.1  | 49.3  | 53.6  | 58.5  | 61.4  | 63.3  | 64.2  | 60.3  | 56.5  | 48.8  | 44.4  | (26) |      |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 48.5  | 52.2  | 56.1  | 59.4  | 63.5  | 67.7  | 70.8  | 71.5  | 66.3  | 62.8  | 56.5  | 51.3 | (27) |
| (28) |                                                                                  | MDCB        | 51.1  | 60.9  | 63.2  | 71.1  | 81.9  | 90.9  | 90.3  | 91.0  | 83.8  | 76.6  | 66.9  | 57.1  | (28) |      |
| (29) |                                                                                  | 2%          | WB    | 46.7  | 49.7  | 53.4  | 57.8  | 62.0  | 66.0  | 68.5  | 69.5  | 64.9  | 61.0  | 54.3  | 49.6 | (29) |
| (30) |                                                                                  | MDCB        | 48.6  | 56.6  | 62.2  | 70.6  | 81.0  | 89.9  | 88.9  | 89.0  | 83.6  | 75.6  | 62.2  | 53.9  | (30) |      |
| (31) |                                                                                  | 5%          | WB    | 45.0  | 47.8  | 51.9  | 56.4  | 61.0  | 64.6  | 67.1  | 68.3  | 63.8  | 59.8  | 52.9  | 48.0 | (31) |
| (32) |                                                                                  | MDCB        | 47.8  | 54.0  | 60.4  | 69.5  | 78.8  | 87.2  | 88.8  | 86.9  | 83.4  | 74.1  | 60.6  | 51.3  | (32) |      |
| (33) |                                                                                  | 10%         | WB    | 43.2  | 46.1  | 50.2  | 55.2  | 59.9  | 63.4  | 65.9  | 67.0  | 62.8  | 58.5  | 51.3  | 46.4 | (33) |
| (34) |                                                                                  | MDCB        | 45.7  | 51.4  | 58.8  | 67.8  | 76.8  | 85.8  | 89.3  | 87.8  | 83.5  | 72.2  | 59.4  | 49.2  | (34) |      |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 12.0  | 14.2  | 16.3  | 18.5  | 21.3  | 23.6  | 23.7  | 23.2  | 21.4  | 18.4  | 16.2  | 12.3  | (35) |      |
| (36) |                                                                                  | 5% DB       | MCDBR | 15.2  | 17.9  | 18.9  | 22.1  | 24.2  | 24.4  | 24.9  | 24.4  | 23.2  | 21.1  | 20.0  | 16.8 | (36) |
| (37) |                                                                                  | MCWBR       | 9.6   | 10.2  | 10.1  | 9.9   | 9.1   | 7.6   | 8.2   | 8.1   | 8.3   | 8.7   | 9.5   | 10.1  | (37) |      |
| (38) |                                                                                  | 5% WB       | MCDBR | 11.4  | 15.2  | 16.4  | 19.4  | 22.1  | 23.9  | 24.2  | 23.9  | 21.5  | 17.9  | 15.3  | 12.0 | (38) |
| (39) |                                                                                  | MCWBR       | 8.1   | 9.1   | 9.6   | 9.7   | 9.0   | 7.8   | 7.9   | 8.1   | 7.9   | 7.8   | 8.3   | 8.1   | (39) |      |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.338 | 0.363 | 0.388 | 0.438 | 0.413 | 0.361 | 0.353 | 0.364 | 0.342 | 0.386 | 0.344 | 0.334 | (40) |      |
| (41) |                                                                                  | taud        | 2.314 | 2.240 | 2.194 | 2.069 | 2.196 | 2.360 | 2.389 | 2.348 | 2.407 | 2.248 | 2.343 | 2.334 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 264   | 272   | 276   | 268   | 276   | 290   | 292   | 286   | 288   | 260   | 260   | 257   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 30    | 37    | 43    | 51    | 46    | 39    | 37    | 38    | 34    | 36    | 29    | 28    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 685   | 945   | 1395  | 1768  | 2160  | 2535  | 2490  | 2207  | 1820  | 1255  | 884   | 647   | (44) |      |
| (45) |                                                                                  | RadStd      | 95    | 135   | 101   | 149   | 134   | 103   | 65    | 69    | 68    | 87    | 89    | 110   | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   | (46) |
|      |                        | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |      |
| (47) | Regional (8 neighbors) | N/A   | N/A     | N/A    | N/A     | -1.71 | N/A   | N/A         | N/A   | N/A   | N/A   | (47) |

Nomenclature: See separate page